In this page,

$C^\infty = \{f : \mathbb{R} \to \mathbb{C} \mid f$ is infinitely differentiable$\}$, Vector space over $\mathbb{C}$.

$\mathbb{R} = C^\infty(I, \mathbb{C})$, where $I \subseteq \mathbb{R}$ is open interval.

$D : C^\infty \to C^\infty : f \mapsto f'$, $\mathbb{C}$-linear map, by linearity of differential

$M_a : C^\infty \to C^\infty : f \mapsto af$, $\mathbb{C}$-linear map, C^∞-module Homomorphism.

$$\begin{aligned} L(y) &= a_n(t) \cdot y^{(n)} + a_{n-1}(t) y^{(n)} + \cdots + a_1(t) y' + a_0(t) y \\ &= (M_{a_n} D^n + M_{a_{n-1}} D^{n-1} + \cdots + M_{a_1} D + M_{a_0}) y. \end{aligned}$$

Prop. $D: \mathcal{B} \to \mathcal{B} : f \mapsto f'$ is linear operator.

Proof. Let $f, g \in \mathcal{B} = C^\infty(I, \mathbb{C})$ and $c \in \mathbb{C}$ be given. Then, for any $x \in I$,

$$D(cf+g)(x) = \lim_{h \to 0} \frac{c \cdot f(x+h) + g(x+h) - (c \cdot f(x) + g(x))}{h}$$

$$= c \cdot \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= c \cdot D(f)(x) + D(g)(x)$$ ☒

Prop. $M_a : \mathcal{B} \to \mathcal{B} : f \mapsto af$ is linear operator for any $a(t) \in \mathcal{B}$.

Proof. Let $f, g \in \mathcal{B} = C^\infty(I, \mathbb{C})$ and $c \in \mathbb{C}$ be given. Then, for any $x \in I$,

$$M_a(cf+g)(x) = a(x) \cdot c \cdot f(x) + a(x) \cdot g(x) = c \cdot M_a(f)(x) + M_a(g)(x).$$ ☒

Prop. For any $a \in \mathcal{B}$,

$$DM_a = M_a D + M_{Da} \qquad (\text{That is, } [D, M_a] = M_{a'}.$$

Proof. Given $f \in \mathcal{B}$,

$$DM_a(f) = D(af) = a'f + af' = M_{a'}(f) + M_a D(f).$$

Thm. On $B = C^\infty(I, \mathbb{C})$,